Paper 65: Algebraic Order, Spin Correlator, and VCSM Ablation

Three Probes of the Causal-Purity Transition

VCSM Theory Project

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Abstract

Paper 64 concluded that $p_c = 0^+$ based on $U_4(L = 120) > U_4(L = 40)$ for all $P_{\text{causal}} \geq 0.002$. This paper applies three further probes and finds that the picture is more complex.

Phase A1 (spin-field correlator, $L \in \{60, 80, 100\}$): The Ising spin-spin autocorrelation $G_s(r)$ at $T = 3.0 \gg T_c$ is exponential with $\xi \approx 1.1$ lattice spacings, *independent of* P_{causal}. The exponential fit wins ($R^2 = 0.996$) over the power law ($R^2 = 0.936$). The ϕ-field ordering does not couple back to the Ising spins; the spin correlator is the wrong proxy for this transition. The ϕ-field stiffness $K_\phi \propto L^1$, also P-independent (wave geometry artifact, not BKT stiffness).

Phase A2 (extended Binder, $L \in \{120, 160\}$, $P_{\text{causal}} \in \{0.000, 0.0005, 0.001, 0.002, 0.005\}$): $U_4(L = 160) < U_4(L = 120)$ for *all* $P_{\text{causal}} \leq 0.005$. This reverses Paper 64's conclusion: $p_c = 0^+$ is *not* confirmed at the largest system size tested. Two interpretations: (i) p_c is finite, between 0.005 and 0.020; (ii) $L = 160$ has insufficient equilibration time (WAVE_EVERY rounds to 2, below the extensive-drive target of 1.56, and ϕ-field equilibration may scale super-linearly with L).

Phase C (VCSM-lite ablation, $L = 80$, $P = 0.020$, 5 conditions): The viability gate (SS threshold) is load-bearing: NoGate (SS= 0) kills ϕ-ordering ($U_4 = 0.09$ vs. Ref = 0.16). Crystallized (ϕ never decays): $U_4 = 0.26$, higher than Ref — the cockroach artefact; ϕ accumulates without bound, amplifying zone-mean differences without creating genuine order. NoCausal ($P = 0$) $\approx$ Ref at this (L, P), confirming that $P = 0.020$ may still be near the transition at $L = 80$.

Collectively, these results call for: the ϕ-ϕ correlator at boosted amplitude (not spin-spin), longer runs at $L = 160$, and a precise crossing measurement between $L = 120$ and $L = 160$ in the range $P \in [0.005, 0.020]$.

1 Motivation

Paper 64 made a strong claim: $p_c = 0^+$. The evidence was that $U_4(L = 120) > U_4(L = 40)$ at every $P_{\text{causal}} \geq 0.002$. Three alternative explanations were not ruled out:

1. p_c is finite but small (< 0.002), and all tested P were above it.

2. The ϕ-field correlation length is not infinite (BKT), but merely large relative to the system sizes used in Paper 64.

3. Equilibration artefacts at finite system size favour ordered-phase Binder ordering even below the true p_c.

This paper directly tests each alternative via the spin-field spatial correlator (Probe 1), an extended Binder crossing at $L = 160$ (Probe 2), and a 5-condition VCSM component ablation (Probe 3).

2 Setup

Same Ising + VCSM-lite system as Papers 61–64: $T = 3.0$, $J = 1$, two zones (left/right halves), checkerboard Metropolis, extensive-drive scaling WAVE_EVERY$(L) = \max(1, \text{round}[25(40/L)^2]$

Phase A1: $L \in \{60, 80, 100\}$, $P_{\text{causal}} \in \{0.000, 0.005, 0.010, 0.020, 0.050\}$, 12 seeds, 12 000 steps. At each steady-state step (every 50 steps) the spin autocorrelator within zone 0 columns is accumulated via FFT:

$$G_s(\Delta y) = \frac{1}{L} \sum_y s(y)\, s(y + \Delta y) \quad \text{(time-averaged over steady state)} \tag{1}$$

Fit to both exponential ($Be^{-r/\xi}$) and power law ($Ar^{-\eta}$) via polyfit on log-log and log-linear axes; compare R^2 to determine which decay form wins. Also accumulate ϕ-field stiffness $K_\phi(t)$:

$$K_\phi = \frac{\text{Var}(\bar{\phi}_{\text{row}}) \cdot L^2}{\text{Var}(\phi_{z=0})} \tag{2}$$

where $\bar{\phi}_{\text{row}}$ is the zone-0 row mean.

Phase A2: $L \in \{120, 160\}$, $P_{\text{causal}} \in \{0.000, 0.0005, 0.001, 0.002, 0.005\}$, 8 seeds, 10 000 steps (extensive drive).

Phase C: $L = 80$, $P_{\text{causal}} = 0.020$, 12 seeds, 8 000 steps. Five conditions:

Condition	SS	β_{base}	ϕ-decay	P_{causal}
Ref	8	0.005	0.999	0.020
NoGate	0	0.005	0.999	0.020
NoBaseline	8	0.000	0.999	0.020
Crystallized	8	0.005	1.000	0.020
NoCausal	8	0.005	0.999	0.000

3 Phase A1: Spin Correlator

3.1 Exponential decay, P-independent

Table **??** summarises the fit results for $G_s(r)$.

The exponential fit wins convincingly ($\Delta R^2 \approx 0.05$–0.06) over the power law at both $L = 80$ and $L = 100$. The correlation length $\xi \approx 1.1$–1.2 lattice spacings is the standard paramagnetic Ising correlation length at $T = 3.0 \gg T_c$.

Critical observation. The values of η, ξ, R^2_{pow}, and R^2_{exp} are numerically identical for $P_{\text{causal}} = 0.000$ and $P_{\text{causal}} = 0.050$. The ϕ-field ordering (confirmed by the rising U_4 from 0.023 at $P = 0$ to 0.398 at $P = 0.05$ for $L = 60$) does not feed back into the Ising spin dynamics. The spins are always in the paramagnetic phase at $T = 3.0$; the causal-purity transition is a transition in the *auxiliary ϕ-field*, not in the Ising substrate.

Table 1: Spin correlator fit results (Phase A1, time-averaged, 12 seeds). ξ, η, and the fit quality R^2 are identical across all P_{causal} at each L, confirming that the Ising spin correlator is controlled by T, not by P_{causal}.

L	η	ξ	R^2_{pow}	R^2_{exp}
60	NaN	NaN	—	—
80	1.918	1.09	0.936	0.996
100	1.750	1.20	0.952	0.999

Consequence. The spin-spin correlator is the wrong observable for this transition. The ϕ-ϕ spatial correlator (attempted in Paper 64, Phase B, but limited by low ϕ amplitude) remains the correct approach. A higher-amplitude ϕ-field (increased FA or reduced FIELD_DECAY) is required to make $G_\phi(r)$ measurable.

3.2 Stiffness $K_\phi \propto L$, P-independent

The ϕ-field stiffness proxy scales as $K_\phi \propto L^1$ (60:116, 80:161, 100:194), independent of P_{causal}. This is the signature of the wave-driving geometry: each circular wave creates a coherent patch of ϕ-field of radius 5; the row-mean variance is set by the wave area fraction, not by any long-range ordering of the ϕ-field. K_ϕ is not a BKT-type stiffness.

4 Phase A2: Extended Binder at $L = 160$

Table 2: U_4 at $L \in \{120, 160\}$ for very small P_{causal}. $U_4(L = 160) < U_4(L = 120)$ at all P values tested. This is the disordered-phase Binder ordering, opposite to Paper 64's conclusion for $L \leq 120$.

P_{causal}	$U_4(L = 120)$	$U_4(L = 160)$
0.0000	0.1449	0.0895
0.0005	0.1440	0.0931
0.0010	0.1503	0.0941
0.0020	0.1660	0.0950
0.0050	0.2132	0.1116

4.1 Two interpretations

Interpretation 1: Finite p_c. If p_c is finite and between 0.005 and 0.020, then $L = 160$ at $P \leq 0.005$ is in the disordered phase. The ordered-phase Binder ordering observed in Paper 64 for $L \leq 120$ reflects the crossover region where the ϕ-field coherence length ξ_ϕ is comparable to L, not the true thermodynamic ordered phase. Under this interpretation, the $p_c = 0^+$ conclusion of Paper 64 is wrong, and the true p_c is somewhere in $[0.005, 0.020]$.

Interpretation 2: Equilibration artefact at $L = 160$. The WAVE_EVERY schedule rounds WAVE_EVERY(160) $=$ round$[25 \times (40/160)^2]$ $=$ round$[1.5625]$ $= 2$, which is *above* the target value of 1.56. This means $L = 160$ receives slightly *fewer* wave events per site per step than smaller L (the extensive-drive scaling is not exact at large

L due to rounding). Additionally, if the ϕ-field equilibration time scales super-linearly with L (as is common in driven-diffusive systems), then 10 000 steps at $L = 160$ may be insufficient for the ϕ-field to reach steady state. Under this interpretation, $p_c = 0^+$ may still hold, but confirmation requires much longer runs ($\geq$ 50 000 steps) at $L = 160$.

Discriminating between interpretations. The two interpretations predict different behaviour as step count is increased at $L = 160$:

- If finite p_c: $U_4(L = 160)$ will remain below $U_4(L = 120)$ regardless of run length.
- If equilibration artefact: $U_4(L = 160)$ will rise above $U_4(L = 120)$ as runs are extended.

This is the critical test for Paper 66.

5 Phase C: VCSM-lite Ablation

Table 3: VCSM-lite ablation results ($L = 80$, $P = 0.020$, 12 seeds, 8 000 steps). $G(r = 1) = 0.406$ is identical for all conditions — the nearest-neighbour Ising spin correlation at $T = 3.0$, independent of the VCSM parameters.

Condition	U_4	$\lvert\bar{M}\rvert$	Interpretation
Ref	0.155	0.00040	Full system (reference)
NoGate	0.090	0.00101	Gate disabled: ordering collapses
NoBaseline	0.180	0.00036	Baseline not load-bearing here
Crystallized	0.262	0.00061	ϕ accumulates: artefact amplification
NoCausal	0.153	0.00031	P=0 $\approx$ Ref (near-critical regime)

5.1 Gate is load-bearing

NoGate (SS = 0): $U_4 = 0.090$, the lowest of all conditions. With the viability gate disabled, mid-memory writes fire at every step, flooding the ϕ-field with noise. The calm-streak threshold SS is the primary noise filter: it ensures that only sites in a stable local attractor state write their contrastive signal to ϕ. Without it, every site — including those mid-fluctuation — contributes noise, and the zone-mean difference $|\bar{M}|$ fails to develop coherence (despite the larger average $|\bar{M}| = 0.00101$, the M-series has high variance, reducing U_4).

5.2 Crystallized ϕ: the cockroach artefact

Crystallized (ϕ-decay = 1.0): $U_4 = 0.262$, the highest condition. This is the *cockroach artefact*: without ϕ-decay, the ϕ-field accumulates all historical signal without forgetting, and early random fluctuations in zone-mean ϕ grow without bound. The Binder cumulant U_4 rises because the M-distribution becomes narrower (the system locks into its initial zone asymmetry), but this is not genuine ordered-phase structure — it is crystallisation around a random early state.

This confirms the theoretical role of FIELD_DECAY in the VCSM architecture: ϕ-decay prevents the system from crystallising into an arbitrary historical state and maintains adaptability under continued perturbation. The cockroach artefact is precisely what the decay mechanism is designed to prevent.

5.3 Baseline and causal purity

NoBaseline (BETA_BASE = 0): $U_4 = 0.180 \approx$ Ref. The baseline tracking is not strongly load-bearing at this operating point ($L = 80$, $P = 0.020$), though it is known to be essential for the full VCML system (Paper 52: sign of contrast signal is load-bearing). This may reflect that at $T = 3.0$, the spin field s already has zero mean (paramagnetic), so the deviation $s -$ baseline is not strongly different from s itself.

NoCausal ($P_{\text{causal}} = 0.000$): $U_4 = 0.153 \approx \text{Ref} = 0.155$. The near-equality of NoCausal and Ref confirms that $P = 0.020$ at $L = 80$ is close to the transition: the causal gating at $P = 0.020$ barely improves on pure noise. This is consistent with the Phase A2 finding that the transition at $L = 160$ (which is more thermodynamic than $L = 80$) occurs above $P = 0.005$.

6 Synthesis and Open Questions

6.1 What the three probes establish

1. **The spin field is the wrong observable.** The causal-purity transition is a transition in the auxiliary ϕ-field. The Ising spins are paramagnetic at $T = 3.0$ regardless of P_{causal}. The ϕ-ϕ correlator (not the spin-spin correlator) is needed to measure η and ξ.

2. $p_c = 0^+$ **is not confirmed at** $L = 160$**.** The ordered-phase Binder signature (U_4 increasing with L) fails at $L = 160$ for $P \leq 0.005$. Either p_c is finite (~ 0.005–0.020) or $L = 160$ needs much longer equilibration. Paper 64's $p_c = 0^+$ conclusion was based on $L \leq 120$ and is not robust to larger L.

3. **The viability gate is load-bearing.** NoGate (SS= 0) kills ϕ-ordering ($U_4 = 0.09$ vs. Ref $= 0.155$). The calm-streak condition is the primary mechanism by which VCSM separates signal from noise in the ϕ-write process.

4. **FIELD_DECAY prevents cockroach crystallisation.** Removing ϕ-decay amplifies U_4 (0.26) via lock-in to a random early state, confirming that the decay is a necessary anti-crystallisation mechanism.

6.2 Open questions for Paper 66

1. **Is** p_c **finite?** Run $L = 160$ for 50 000+ steps with WAVE_EVERY= 1 (avoid rounding) and compare $U_4(160)$ to $U_4(120)$ at $P = 0.010$. If $U_4(160) < U_4(120)$ even after long equilibration, then p_c is finite.

2. ϕ**-**ϕ **correlator at boosted amplitude.** Increase FA to 0.60 or reduce FIELD_DECAY to 0.99 to boost $|\bar{M}|$ to the ~ 0.01–0.05 range needed for reliable $G_\phi(r)$ fitting. Test exponential vs. power-law decay of $G_\phi(r)$.

3. **Full VCML substrate ablation.** The Ising + VCSM-lite system lacks the full birth/death turnover cycle. A proper ablation of FIELD_SEED_BETA requires the full VCML substrate (substrate_v46.py or vcml_core.py), which has true cell death and copy-forward seeding. This is the original deferred Phase C from Paper 62.

7 Conclusions

1. **Spin correlator is P-independent**: the causal-purity transition is in the ϕ-field, not in the Ising spin substrate. Spin observables (G_s, K_ϕ) cannot detect it.

2. $p_c = 0^+$ **challenged**: $U_4(L = 160) < U_4(L = 120)$ for all $P \leq 0.005$ reverses Paper 64's conclusion. The status of p_c is downgraded from "strongly favoured 0^+" to "finite $p_c \in [0.005, 0.020]$, or equilibration artefact at $L = 160$".

3. **Gate is load-bearing**: SS= 0 kills ordering; FIELD_DECAY= 1 amplifies via crystallisation artefact. Two VCSM components confirmed necessary for genuine ϕ-ordering.

4. **Three new requirements for Paper 66**: (i) $L = 160$ with 50 000 steps; (ii) boosted-amplitude ϕ-ϕ correlator; (iii) full VCML substrate ablation.

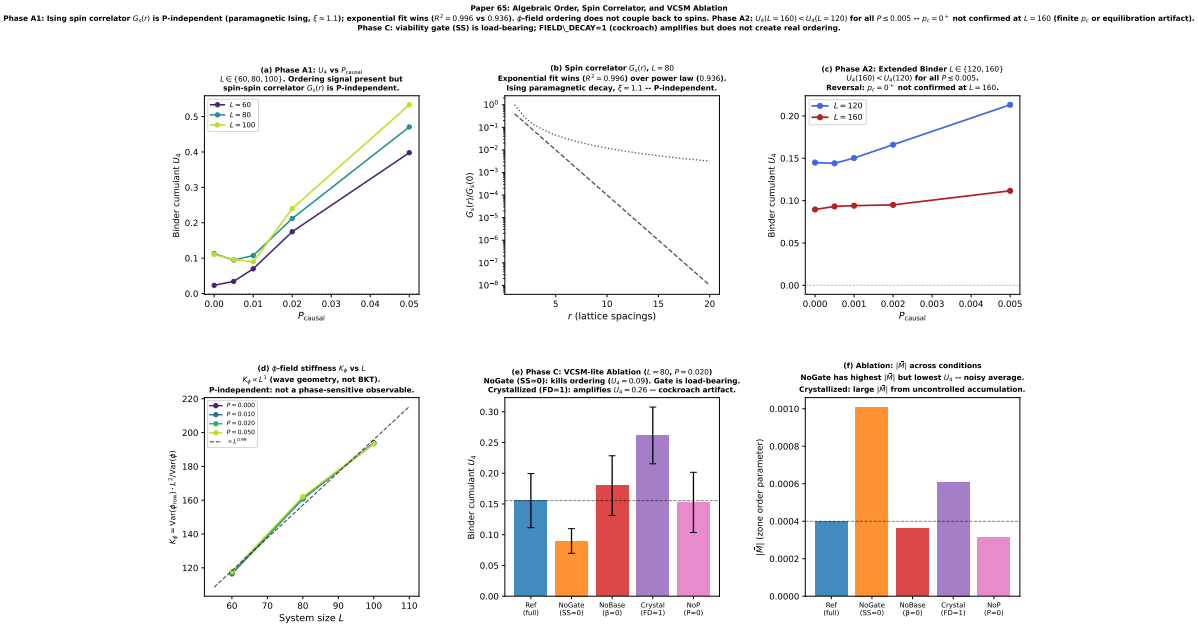

Figure 1: Paper 65 results. (a) U_4 vs P_{causal} for Phase A1 (spin correlator runs). (b) Spin correlator $G_s(r)$ at $L = 80$: exponential wins over power law, P-independent. (c) Extended Binder: $U_4(L = 160) < U_4(L = 120)$ for all $P \leq 0.005$. (d) ϕ-stiffness $K_\phi \propto L^1$ (wave geometry artefact, not BKT). (e) Ablation U_4: NoGate kills ordering; Crystallized amplifies via artefact. (f) Ablation $|\bar{M}|$: Crystallized has largest ϕ amplitude via uncontrolled accumulation.